

In vitro anti-hyperglycemia properties of the aqueous stem bark extract from *Strychnos henningsii* (Gilg)

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Abstract *Strychnos henningsii* (SH) is a plant commonly used in southern African traditional medicine for the management of diabetes mellitus. Previous in vivo studies showed that a stem bark extract improves glycemic control in a diabetic animal model. Meanwhile, the mechanism of action has not been elucidated. The present study therefore investigated various in vitro models known to target glucose homeostasis and its direct complications. The plant extract was found to stimulate both basal and insulin stimulated glucose uptake in differentiated 3T3-L1 cells but not in Chang liver cells. The effect on 3T3-L1 cells appears independent of PPAR γ as the extract did not stimulate adipogenesis. Although, SH extract was inhibitory toward intestinal α glucosidase, the physiological relevance is doubtful based on the recommended dosages. The extract strongly inhibited protein glycation which, at least in part, may be explained by the antioxidant and phenolic content of this plant. Cytotoxicity in Chang liver cells yielded an IC₅₀ value of 130.0 μ g/mL raising concern that continual exposure to this herbal remedy may initiate hepatotoxicity. The finding from this study suggests that SH extract may attenuate hyperglycemia through enhanced peripheral tissue glucose utilization.

Keywords *Strychnos henningsii* · Hyperglycemia · Hepatotoxicity · Glucose uptake · Adipocyte · Protein glycation

Introduction

Diabetes mellitus is a heterogeneous group of metabolic disorders characterized by hyperglycemia and glucose intolerance as a result of inadequate supply of insulin and/or impaired effectiveness of insulin to stimulate glucose uptake. Persistent hyperglycemia leads to further complications including non enzymatic glycation of proteins, hyperlipidemia and oxidative stress, all of which are believed to be pathogenic in the development of overt diabetes [1–3]. Given that diabetes is a multifactorial chronic disease, numerous therapeutic targets can be exploited in its management as evident from the variety of drugs used clinically. However, the currently available antidiabetic drugs, with perhaps the exception of metformin, remain largely unaffordable to the majority of patients in developing African countries. Consequently, these patients frequently consult traditional healers who are heavily dependent upon medicinal plants for the management of diabetes mellitus [4]. Some of these plants are claimed to possess antidiabetic properties due to a wide range of phytochemical constituents with known antidiabetic properties and may therefore represent a more cost effective treatment, often with lesser side effects [5]. Unfortunately, the majority of plants used in southern Africa as traditional medicine are yet to be thoroughly explored to define the safety and efficacy underlying their potential as complementary or alternative treatment for diabetes and its related complications.

Strychnos henningsii Gilg (Loganiaceae) is one of the plants used in southern and eastern Africa as a folk remedy for the management of various diseases [6]. A decoction of the plant has been used in traditional Kenyan medicine for the treatment

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of rheumatism, gynaecological complaints, chest pain, internal injuries and malaria [7]. It is also used in Tanzania for the treatment of snake bite and hookworm infections. This folklore remedy was discovered recently during our ethnobotanical survey in Nkonkobe Municipality, Eastern Cape, South Africa as herbal treatment for the management of diabetes mellitus [6]. In South Africa, the aqueous bark extract of the plant is also used traditionally for the treatment of stomach ache, colic, dizziness and as a purgative agent. Some compounds such as holstiine, diaboline, strychnochromine and guianensine have been isolated from the *Strychnos henningsii* root and stem barks [8]. Previously, we have demonstrated the in vivo antidiabetic effects of an aqueous stem bark extract of SH on the fasting blood glucose and glucose tolerance in a diabetic rat model [9]. The present study was therefore undertaken to further investigate this antidiabetic mechanism by using various in vitro models known to target glucose homeostasis and its direct complications, including hepatocyte and adipocyte glucose utilization, intestinal carbohydrate digestion, oxidative stress and non-enzymatic protein glycation.

Experimental section

Plant material

The stems bark of SH was collected from a thick forest in February, 2009 through a traditional healer in the Amathole District (Eastern Cape, South Africa). The plant was identified by its vernacular name and later authenticated by Prof. D.S. Grierson of Botany Department, University of Fort Hare. Voucher specimen (Sun MED 2009) was deposited at the Giffen herbarium of the University. The herbal remedy is normally prepared by soaking 3 g of the plant sample in 750 mL of water and ½ of a cup (150 mL) is taken orally three times a day for the treatment of a diabetic patient [6].

Preparation of bark extract

The stem bark material was air-dried at room temperature in the laboratory and further pulverized into powdery form using an electric blender (Waring Products Division, Torrington, USA). Approximately 80 g of the powdered plant material was macerated in 1,000 ml of cold distilled water (4 °C) maintained on a mechanical shaker (Stuart Scientific Orbital Shaker, SOS1, Essex, UK) for 48 h. The extract was filtered through Whatman No.1 filter paper using a Buchner funnel. The filtrate was immediately frozen at −40 °C and lyophilized for 48 h using a freeze dryer (Savant Refrigerated vapor Trap, RV T41404, USA) to give a yield of 10.6 g of dry extract. The dried extract was reconstituted in DMSO, vortexed and left for at least 15 min prior to further dilution with the respective tissue culture medium or assay buffer.

Cell lines, media and chemicals

Chang liver and 3T3-L1 cell lines were obtained from Highveld Biological, (Johannesburg, South Africa) and ATCC respectively. Glucose oxidase reagent, DMEM and RPMI 1640 medium were obtained from Sigma Aldrich, South Africa, whereas fetal calf serum (FCS) was purchased from Lonza Biologics. All other reagents used in this study were of analytical grade and were purchased from Sigma or Merck Chemicals (Pty) Ltd, South Africa.

Maintenance of cell cultures

All the cell cultures were incubated at 37 °C in a humidified atmosphere with 5 % CO₂. The Chang liver cells were replenished with new growth medium every 2–3 days, consisting of RPMI 1640 medium supplemented with 10 % fetal bovine serum. 3T3-L1 cells were cultured in DMEM with 10 % fetal bovine serum. Both cell lines were sub-cultured before 80 % confluence was reached.

Glucose utilization in 3T3-L1 cell line

For glucose utilization determination, 3T3-L1 preadipocytes were seeded at a density of 3000 cells per well into a 96-well plate. Two days after reaching confluence the cells were induced to differentiate by treating with 1 μM Rosiglitazone and cultured for an additional 5 days by which time the cells accumulated substantial lipid droplets. The method described by van de Venter et al. [10] was adopted to determine the glucose uptake. Forty eight hours prior to glucose utilization assay the plant extract was added and the incubation continued for a further 48 h. Spent culture medium was removed and replaced with fresh RPMI medium, diluted to 8 mM glucose with phosphate buffered saline (PBS), containing 0.1 % bovine serum albumin (BSA) with or without insulin and incubated for 1.5 h. Five microlitre of the assay medium was then removed and used to quantify the remaining glucose using a glucose oxidase based assay kit (SERA-PAK Plus, Bayer). The amount of glucose utilized was calculated by subtracting the amount of glucose left in the medium from the amount present at time 0 (8 mM). Toxicity assays were conducted simultaneously by adding MTT (0.5 mg/mL in RPMI1640) to the wells of the last three rows of each plate and the absorbance measured at 560 nm after solubilising the formazan crystals with DMSO.

Glucose utilization in Chang liver cells

The Chang-liver cells were cultured and maintained in an antibiotic-free growth medium as described above. The cultured cells were seeded into 96 well plates at a density of 6000 cells per well in growth medium (RPMI 1640 medium

supplemented with 10 % fetal bovine serum) for 3 days without medium change, prior to glucose utilization assay. On day three after seeding, SH extract and or metformin was added to give the final concentration of 12.5 µg/well of the extract. Forty eight hours after treatment, the medium was removed and 50 µL of incubation medium (8 mM glucose RPMI 1640+0.1 % BSA) containing the plant extract (12.5 µg/mL) was added to the appropriate wells. 1 µM metformin was used as the positive control while the control contained only incubation medium. After 180 min of incubation, 5 µL was removed and placed into a new 96 well plate into which 200 µL glucose oxidase reagent (SERA-PAK Plus, Bayer) was added. After incubation for 15 min at 37 °C, the absorbance was measured at 510 nm using a Multiscan MS microtiter plate reader (Lab systems). The amount of glucose utilized was calculated by subtracting the amount of glucose left in the medium from the amount given at time 0. Toxicity assays were conducted simultaneously by adding MTT (0.5 mg/mL in RPMI1640) to the wells of the last three rows of each plate and the absorbance measured at 560 nm after solubilising the formazan crystals with DMSO.

Cytotoxicity in Chang liver cells

Chang liver cells were seeded into 96-well plates at a density of 8000 cells per well and allowed to reach full confluence. Cells were treated with the indicated concentrations of plant extract for 48 h after which the cells were incubated with 0.5 mg/ml MTT at 37 °C for an additional 3 h. The supernatants were discarded and the formazan crystals dissolved in DMSO. After incubating at room temperature for 20 min the absorbance was measured at 570 nm using a microplate reader. The IC₅₀ value was determined using Graph Pad Prism software.

Triglyceride accumulation in preadipocytes

Two day post confluent 3T3-L1 cells were treated with the indicated concentration of plant extract in fresh medium and cultured for an additional 7 days. The cells were fixed by replacing growth medium with 100 µL of 4 % formaldehyde in PBS. After fixing the cells for 1 h at room temperature, the formaldehyde solution was removed and the cells were washed twice with 200 µL of PBS. A volume of 100 µL crystal violet (0.1 % w/v in distilled water) was added to each well for 30 min at room temperature. Excess dye was removed by washing the plates with water and later dried the plates in the oven at 37 °C. A volume of 50 µL of oil red O (0.5%w/v prepared in isopropanol) was added to each well and incubated at 37 °C for 15 min. The stained cells were washed with acetic acid (100 µL; 10%v/v). Triglyceride accumulation was quantified by measuring the absorbance at 595 nm. After Oil Red-O staining cells were stained with crystal violet to normalize

cell density. Rosiglitazone (1 µM) was used as a positive inducer of adipogenesis.

Inhibition of protein glycation

Albumin glycation was determined using the method of Matsuura et al. [11]. Six tubes were prepared for the blank and each of the extract concentrations to be tested by mixing 400 µL of 1 mg/mL BSA with 90 µL of 1.11 M glucose in 50 mM phosphate buffer (pH 7.4). Three of these samples were incubated in a 60 °C waterbath for 24 h with 10 µL of extract (3, 6 and 12.5 µg/mL) or distilled water while another three were kept at 4 °C. After incubation, 100 µL of each sample was transferred to a new tube and 10 µL of cold 100 % TCA was added to each tube. They were vortexed and centrifuged at 15000 × g for 4 min at 4 °C. The pellet [containing advanced glycation end products, (AGE)] was collected and dissolved in 400 µL of alkaline PBS. The fluorescence intensity was measured using a Multiscan MS microtiter plate reader with an excitation/emission wavelength of 370/440 nm.

Alpha-glucosidase inhibitory activity

Alpha glucosidase solution was prepared from rat intestinal acetone powder. Exactly 100 mg of acetone powder was suspended in 3 mL of 0.01 M phosphate buffer (pH 7.0) and sonicated 12 times for 30 s each in an ice bath followed by centrifugation at 3,000 rpm, 4 °C for 20 min. The supernatant containing the enzyme was obtained and kept on ice. Serial dilutions of this solution were assayed to determine the enzyme yield and further diluted to approximately 0.1 U/mL with 0.01 M phosphate buffer (pH 7.0). The inhibitory activity of alpha-glucosidase was evaluated according to the method previously reported by Sancheti et al. [12]. The reaction mixture consisted of 50 µL of 0.5 mM MPNG (dissolved in 0.1 M phosphate buffer pH 7.0), 10 µL of test sample and 25 µL of glucosidase solution prepared from the stock solution (1 mg/mL in 0.01 M phosphate buffer pH 7.0). The reaction mixture was then incubated at 37 °C for 30 min and terminated by the addition of 100 µL of 0.2 M sodium carbonate solution. The absorbance was measured at 410 nm using Multiscan MS microtiter plate reader. The % inhibition was calculated as follows: $(1 - B/A) \times 100$ where A is the absorbance of control and B is the absorbance in the presence of plant extract. All experiments were carried out in triplicate and the results are expressed as the mean ± SD.

Alpha- amylase inhibitory activity

The alpha-amylase inhibitory activity of SH extract was tested based on the method described by Zhizhuang et al. [13] using microplate-based starch-iodine assay. The

reaction was initiated by adding 40 μL of starch (Sigma EC 232) solution (2 g/L) and 40 μL of 20 μg porcine pancreatin (Sigma A; 3176-1MU) in 0.1 M phosphate buffer at pH 7.0 to microplate wells. After 30 min of incubation at 50 $^{\circ}\text{C}$, 20 μL of 1 M HCl was added to stop the enzymatic reaction. 100 μL of iodine reagent (5 mM I_2 and 5 mM KI) was added to the mixture. After colour development, 150 μL of iodine treated sample was transferred to a transparent flat bottomed 96 well microplate to measure the absorbance at 580 nm.

Total phenolic content

The amount of phenolic compounds in the aqueous stem bark extract of SH was determined with Folin Ciocalteu reagent using the method of Lister and Wilson [14]. To 0.5 mL of plant extract solution (1 mg/mL), 2.5 mL of 10 % Folin-Ciocalteu reagent and 2 mL of Na_2CO_3 (2 % w/v) were added. The resulting mixture was incubated at 45 $^{\circ}\text{C}$ with shaking for 15 min and the absorbance at 765 nm was measured. Total phenolics content was expressed as mg/g tannic acid equivalent using the following equation from the calibration curve: $Y=0.1216x$, $R^2=0.936512$, where x was the absorbance and Y was the tannic acid equivalent (mg/g). The experiment was conducted in triplicate, and the results are reported as mean $\pm$ SD values.

Ferric reducing antioxidant potential (FRAP)

The method described by Benzie and Strain [15] was used to determine total antioxidant activity of SH extract. The FRAP reagent was prepared by mixing 20 mL acetate buffer (300 mM, pH 3.6), 2 mL TPTZ solution, with 2 mL FeCl_3 solution and 2.4 mL of distilled water. The straw colored solution was incubated in a water bath at 37 $^{\circ}\text{C}$ for 10 min. A volume of 10 μL of the sample was transferred into a 96 well plate with 250 μL of freshly prepared FRAP reagent. After 5 min, the absorbance was measured and recorded at 593 nm against blank solution (distilled water instead of extract). The antioxidant capacity of the extract based on the ability to reduce ferric ions was expressed as catechin equivalents (mg/g) using the following equation from the calibration curve: $Y=0.3329x$, $R^2=0.9763$ where x was the absorbance and Y the catechin equivalent (mg/g).

Statistics

Data analysis was done on Microsoft Excel to obtain descriptive statistics. Means were separated by the Duncan multiple test using SAS. The different levels of significance within the separated groups were analyzed using one way analysis of variance (ANOVA). Values were considered significant at $P<0.05$.

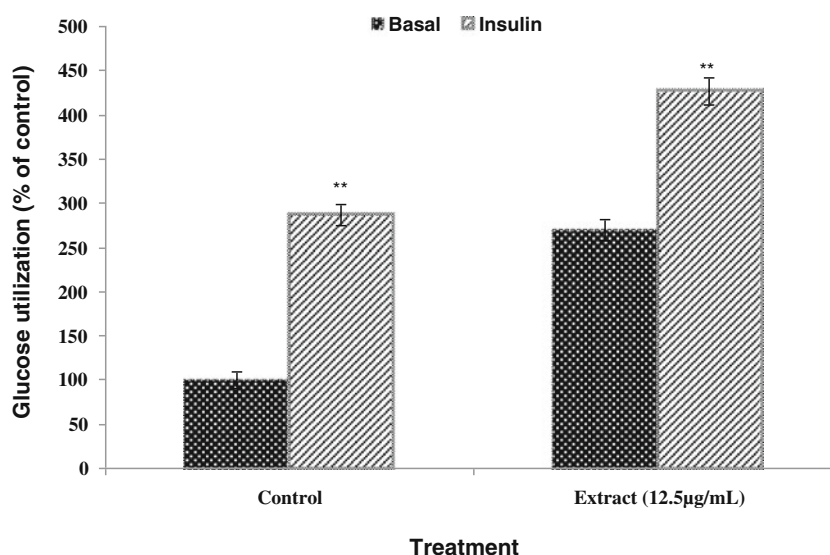
Results and discussion

Type 2 diabetes is a complex metabolic disorder caused by abnormal insulin secretion or insulin action resulting from impaired β -cell function and insulin resistance in the liver, skeletal and adipose tissues [16]. It is characterized by chronic high blood glucose that causes glycation of body proteins [17] and oxidative damage especially in vulnerable pancreatic β -cells [18]. While the precise underlying mechanism responsible for this disease and its related complications remain to be fully elucidated, defective insulin signaling is considered to be one of the primary events leading to the development of hyperglycemia. Previous animal studies have documented that extracts from SH have the capability to reduce glycemia in animal model [9]. To date the main mechanisms recognized to contribute to the therapeutic maintenance of normal glucose levels can be attributed to the intestinal inhibition of glucose uptake, decreased hepatic glucose output and the enhanced clearance of blood glucose through altered uptake by peripheral tissues. Increasing evidence indicates that persistent hyperglycemia is the initiating cause of the tissue damage occurring in diabetes, both repeated acute changes in cellular glucose metabolism or long-term accumulation of glycated biomolecules, advanced glycation end products and related oxidative stress [19].

Glucose uptake in differentiated 3T3-L1 and Chang liver cells

Treatment of differentiated 3T3-L1 cells with the extract produced a greater than two fold increase in the rate of basal glucose utilization (Fig. 1). In the presence of insulin a further increase was evident; however the effect was substantially less as would be expected from synergistic or complete additive effect. In contrast to adipocytes, treatment of Chang liver cells revealed no significant enhancement in glucose uptake (Fig. 2). The increased glucose utilization observed in differentiated 3T3-L1 cells treated with SH extract both under basal and insulin stimulated conditions suggest that the extract may reduce hyperglycemia via mechanism involving peripheral tissue glucose uptake. It is now well established that translocation of the glucose transporter GLUT4 to the plasma membrane is a pre-requisite for the stimulation of glucose uptake by insulin in both muscle and adipose tissues. In contrast to adipose tissue, liver glucose uptake is independent of insulin and of the expression and function of the GLUT4 transporter. This difference in the insulin sensitivity of glucose uptake by adipose and hepatocyte are mirrored by the effects of SH extract on these cell types and support a molecular mechanism in which the plant extract may target the GLUT4 insulin signaling cascade [20]. The binding of insulin to the insulin receptor provides docking

Fig. 1 The effect of aqueous extract of *Strychnos henningsii* bark on glucose utilization in differentiated 3T3-L1 cells. Cells were exposed to extract concentration at 12.5 $\mu\text{g/mL}$ for 48 h prior to the glucose utilization assay and to insulin (1.0 μM) during the 90 min glucose utilization assay. All glucose utilization data was normalized for differences in cell numbers using MTT data from parallel treated wells. Data are represented as % of control $\pm$ SD of mean ($N=6$). $**P<0.01$ compared to insulin treatment



site for several downstream signaling molecules that transduce to activation of phosphatidylinositol -3- kinase (PI-3-K) [15]. PI-3-kinase is an important step in the insulin signaling to glucose transport through GLUT4 translocation. Increase in enzyme activity of PI-3-kinase enhances the recruitment of Protein Kinase C (Akt) which promotes translocation of GLUT1 and GLUT4 to the plasma membrane [20–22], facilitating the uptake of glucose from the bloodstream into the cells [23]. Although this pathway provides numerous potential sites through which SH extract may alter glucose metabolism, however, we do not discard the possibility of an additional insulin independent mechanism. We observed that simultaneous treatment with both plant extract and insulin leads to some degree of addition, but it did not reach the sum of both

independent effects. Our observation suggests that multiple sites are not targeted to the enhanced glucose utilization in the adipocytes. Further studies are needed to determine the precise molecular target by which SH increases the adipocyte glucose uptake. This folk medicine did not stimulate adipocyte differentiation in 3T3-L1 cells indicating that the effects are most likely independent of PPAR γ (Fig. 3).

Metformin is a popular drug prescribed for the treatment of type 2 diabetes however the precise molecular mechanism remains poorly defined [24]. The glucose lowering effects associated with metformin results mainly from decreased hepatic glucose production via the inhibition of gluconeogenesis and to a lesser extent to increased insulin stimulated glucose utilization in peripheral tissue [25]. It has been shown to inhibit the mitochondrial respiratory chain at complex I with concomitant up-regulation of glucose transporters and glycolytic enzymes resulting in increased glycolytic glucose utilization hence used as positive control for the

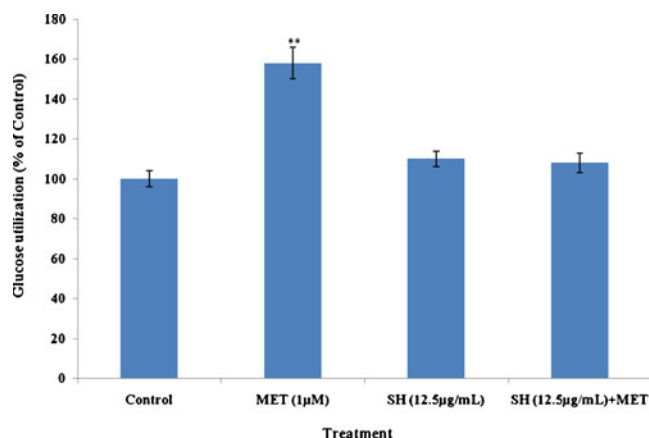


Fig. 2 The effect of aqueous stem bark extract of *Strychnos henningsii* on glucose utilization in Chang liver cells. Cells were exposed to extract concentration of 12.5 $\mu\text{g/mL}$, metformin (MET) and combination of extract and metformin (SH + MET) 48 h prior to glucose utilization assay. All glucose utilization data was normalized for differences in cell numbers using MTT data from parallel treated wells. Data are expressed as % of control $\pm$ SD ($N=6$). $**P<0.01$ compared to the control and extract treatment

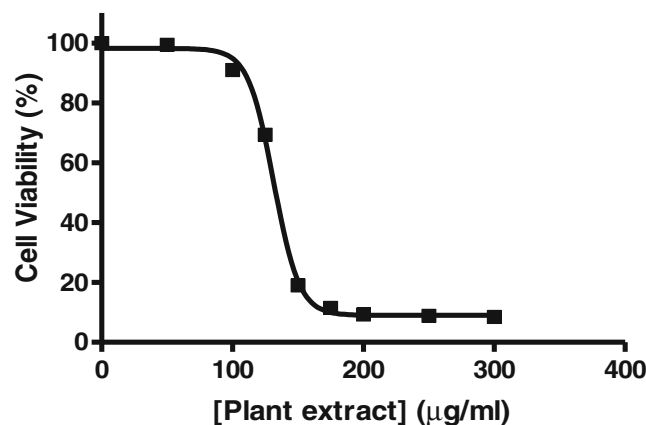


Fig. 3 MTT viability assay of Chang liver cells treated with aqueous extract of *Strychnos henningsii*. Confluent cultures were treated for 48 h prior to MTT assay. Data are expressed as % of control $\pm$ SD ($N=6$). The IC_{50} of the extract is 131.5 ± 6.1 $\mu\text{g/mL}$

Chang liver assay [26]. Metformin exhibited higher glucose uptake activity with response of 56.74 % over the extract (Fig. 2). The plant extract appeared to inhibit the action of metformin when simultaneously added to Chang liver cells, indicating that SH extract may antagonize the clinical efficacy of metformin when co-administered and raises concern of therapeutic failure. Our data therefore provided safety precaution to the majority of people especially from rural communities who used herb-drug combination for the treatment of diabetes.

MTT toxicity assay

The cytotoxicity results obtained from MTT toxicity assay demonstrate that SH extract was not significantly toxic to the Chang liver cells at the concentrations used to measure glucose utilization (12.5 µg/mL). However, higher concentrations led to significant toxicity, with an IC₅₀ value of 130 µg/mL as shown in Fig. 3. Considering that diabetes is a chronic disease requiring constant therapeutic intervention, incessant administration of SH may bring about serious side effects such as hepatotoxicity. Although the genus *Strychnos* is very well known for the production of strychnine [27], the occurrence of strychnine appears to be quite rare in *Strychnos* species. To date, strychnine has been found in only one African species, namely *S. icaja* Baill [28]. Furthermore, roots are the known site of biosynthesis of strychnine and consequently root bark extracts have a greater pharmacological activity than other plant parts in the *Strychnos* species [29]. The strychnine derivative diaboline found in SH is considerably less toxic due to the opening of the lactam of the basic strychnine scaffold and the presence of a hydroxyl group at the 17-position, as a result even high doses of this compound were reported to have a weak curare effect [30].

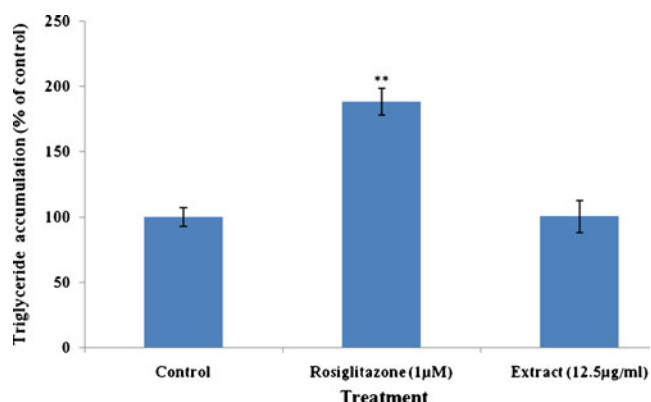


Fig. 4 Effect of *Strychnos henningsii* extract on 3T3-L1 preadipocyte triglyceride accumulation after incubation for 15 min at 37 °C. Lipid accumulation in the cells treated with extract was 100.31±2.21 and that of Rosiglitazone was 188.10±3.15. Results were corrected to cell density and represent the mean ± SD of 4 replicates. ***P*<0.01 compared to the control and extract treatment

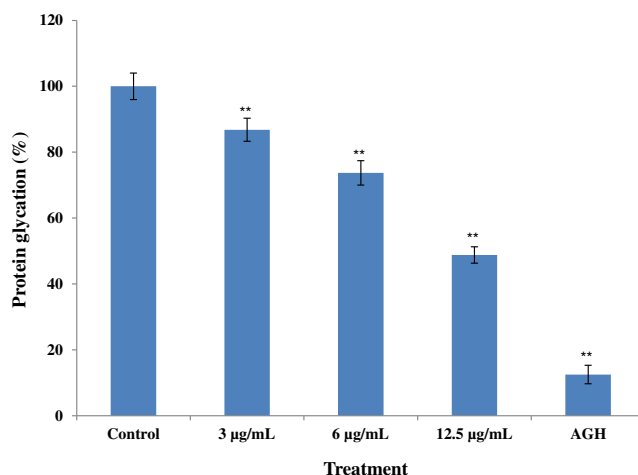


Fig. 5 Effect of *Strychnos henningsii* extracts and aminoguanidine (AGH) on BSA glycation. Data are expressed as mean ± SD (*N*=6). ***P*<0.01 compared to control

Lipid accumulation assay

The effect of plant extract on lipid accumulation during differentiation of preadipocytes of 3T3-L1 cells to adipocytes is examined using oil-red-O stained cells (Fig. 4). Plant extract did not have any significant effect on the lipid accumulation as compared with insulin treated cells at physiological concentration (12.5 µg/ml). The observed results give an indication that the extract might not be a good therapeutic agent in lowering lipid profiles associated with obesity especially in the in vitro studies. Rosiglitazone is a standard antidiabetic drug used as an insulin sensitizer and making the cells more responsive to insulin.

Protein glycation

Protein glycation is one of the consequences of elevated blood glucose in diabetic patients [31]. This is caused by the non - enzymatic reaction between reducing sugar such as glucose and free amino group of proteins to form Amadori products [32]. The rearrangement of this product either by oxidation or reduction further leads to the formation of

Table 1 Total antioxidant (FRAP) activity and phenolic content of the bark extract of *Strychnos henningsii* and some standard compounds

Sample	FRAP value
Extract	357.05±9.21 ^a
Catechin	859.32±15.67 ^a
BHT	75.22±5.16 ^a
Quercetin	2907.25±23.67 ^a
Total phenolics content (extract)	48.02±1.52 ^b

^a Expressed in units of µmol Fe (II)/g

^b Expressed as Tannic acid equivalents (mg/g)

advanced glycation end-products (AGEs). This product has been reported to cause various complications in patients with diabetes due to loss of collagen [32]. The plant extract suppressed the intensity of the fluorescent conjugated BSA (advanced glycation products) in a concentration dependent manner (Fig. 5). The percentage protein glycation in the medium was reduced to 86.8, 73.7 and 48.8 % at 3, 6 and 12.5 $\mu\text{g/mL}$ respectively when compared with the control while that of aminoguanidine, a known inhibitor of protein glycation process, was reduced to 12.5 %. The result obtained demonstrated that SH extract has anti-glycation activity at a concentration (12.5 $\mu\text{g/mL}$) which is considered physiologically relevant.

Alpha glucosidase and alpha-amylase inhibition

Small intestine alpha glucosidase and alpha amylase are key enzymes of dietary carbohydrate digestion in humans. One of the important therapeutic approaches to decrease postprandial glucose is to retard absorption of glucose through inhibition of carbohydrate hydrolyzing enzymes such as alpha amylase and alpha glucosidase [33]. Alpha amylase degrades complex dietary carbohydrates to oligosaccharides and disaccharides, which are ultimately converted into monosaccharides by alpha glucosidase. In vitro alpha glucosidase and alpha amylase inhibitory activities showed that SH extract had an appreciable inhibitory effect on alpha glucosidase but weak inhibitory effect on alpha amylase. The percentage inhibitory activity of the extract against alpha glucosidase at 0.25, 0.5 and 1.0 mg/mL was 14.8, 35.27 and 58.04 % respectively but had weak inhibitory potential against alpha amylase. The observed results indicate that SH extract probably functions through other mechanisms as anti-diabetic herbal remedy and not through inhibition of these digestive enzymes. Although the extract demonstrated an in vitro inhibitory effect on rat intestinal alpha-glucosidase, this inhibition has doubtful physiological relevance. Based on the percentage yield and recommended dosage it can be estimated that the quantity of plant extract taken per session is approximately 60 mg. Considering that the average gut volume for a 70 kg human has been estimated to be 500 mL, the final gut concentration of the plant extract at the dosage recommended traditionally would be in the region of 60 $\text{mg}/500\text{ mL}$ which is equivalent to 0.12 mg/mL . Therefore, at the expected gut concentration the inhibitory activity will be negligible.

Ferric reducing antioxidant power assay (FRAP)

Since antioxidant activity may have contributed to the anti-glycation effects observed in this study, FRAP and total phenolics contents were determined. Phenolics have been documented to possess significant antioxidant and free radical scavenging activity which may be responsible for the

reduction of TPRZ-Fe (III) complex to TPTZ-Fe (II) as shown in this study. The results obtained depicted high antioxidant activity of the plant extract (357.05 $\mu\text{mol Fe (II)/g}$) when compared with BHT (75 $\mu\text{mol Fe (II)/g}$) but significantly lower than that of catechin (859.32 $\mu\text{mol Fe (II)/g}$) and quercetin (2907.25 $\mu\text{mol Fe (II)/g}$) as depicted in Table 1. The value of phenolic content showed good correlation with the antioxidative nature of this folk medicine. The antioxidant potential of SH extracts could play a major role in inhibiting protein glycation due to the presence of phenolic compounds which have been reported to prevent AGE formation [34].

Conclusions

In summary, the data obtained from this study suggests that enhanced adipose tissue glucose uptake represents a mechanism through which the aqueous stem bark extracts of SH improves glycemic control. Toxicity and the antagonistic effect with metformin raises concern as to the safety of this plant in the treatment of diabetes. If SH directly targets the insulin signaling cascade then it is probable that muscle tissue glucose uptake, which also depends on GLUT4 cascade, will be similarly affected. Further studies on the effect of SH on insulin signaling pathways are needed to define the specific mechanism of action.

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Conflicts of interest None

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